SMART ROOM ACTIVITY MONITORING AND BEHAVIOR ANALYSIS USING ESP32 AND MACHINE LEARNING

MINOR PROJECT REPORT

Submitted in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

Department of Computer Science and Engineering

Submitted to

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL (M.P.)

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April 2026

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL (M.P.)

CERTIFICATE

This is to certify that the work embodied in this report entitled **"Smart Room Activity Monitoring and Behavior Analysis using ESP32 and Machine Learning"** has been satisfactorily completed by **Hariom Yadav (23U02083) and Sardul V Pandey (23U02092).** It is a bonafide piece of work, carried out under our guidance in the **Department of Computer Science and Engineering**, **Indian Institute of Information Technology, Bhopal** for the partial fulfillment of the Bachelor of Engineering during the academic year 2025-26.

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DECLARATION

We hereby declare that the following major project synopsis entitled "Smart Room Activity Monitoring and Behavior Analysis using ESP32 and Machine Learning" presented here is in the partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in** Computer Science and Engineering. It is an authentic documentation of our original work carried out under the able guidance of **Dr. Priyankar Choudhary**. The work has been carried out entirely at the Indian Institute of Information Technology, Bhopal. The project work presented has not been submitted in part or whole to award of any degree or professional diploma in any other institute or organization.

We, with this, declare that the facts mentioned above are true to the best of our knowledge. In case of any unlikely discrepancy that may occur, we will be the ones to take responsibility.

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Sardul V Pandey (23U02092)

ACKNOWLEDGEMENT

We, Hariom Yadav and Sardul V Pandey, would like to express our sincere gratitude to everyone who supported and guided us throughout the successful completion of our minor project titled "Human Activity Recognition with AI-Based Health Reporting".

First and foremost, we would like to extend our heartfelt thanks to our esteemed **Director**, **Prof. Ashutosh Kumar Singh**, for providing us with the necessary resources and facilities for conducting our project. His encouragement and vision played a key role in helping us accomplish our goals.

We are deeply grateful to our **Project Guide**, **Dr. Priyankar Choudhary**, for his/her constant guidance, invaluable suggestions, and expert advice throughout the project. His/her knowledge and insights helped us enhance the quality of our work, and his/her support kept us motivated during difficult times.

Our sincere thanks to the **Department Coordinator**, **Dr. Rahul Chaurasia** ,for organizing and overseeing the entire project process. His/her assistance in clarifying doubts and providing us with the right direction was instrumental in the smooth execution of our project.

We would also like to express our gratitude to all the **faculties** of the CSE for their continuous support and encouragement. Their teaching and knowledge were invaluable in enhancing our technical skills and understanding, which helped us in the successful completion of the project.

A special thanks to our **friends** for their continuous support, valuable suggestions, and helping us during various stages of the project. Their insights and encouragement motivated us to keep working on the project with enthusiasm.

Last but certainly not the least, we would like to express our deepest gratitude to our **parents** and family members for their love, patience, and continuous moral support throughout the course of this project. Their understanding and encouragement were the driving force behind our efforts.

We are truly thankful to everyone who directly or indirectly contributed to the success of this project. Without their assistance and support, this project would not have been possible.

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AREA OF WORK

This project is mainly based on the integration of **Internet of Things (IoT), embedded systems, and machine learning** to monitor and analyze human activities in a real-world environment. The primary focus is on collecting real-time data using multiple sensors connected to an ESP32 microcontroller and using that data to understand human behavior patterns.

Unlike traditional systems that depend on cameras or wearable devices, this project uses non-invasive sensors such as motion sensors, light sensors, temperature and humidity sensors, and distance sensors. These sensors continuously capture environmental and positional data without disturbing the user. The ESP32 acts as the central unit that processes this sensor data and sends it to a cloud platform using WiFi.

A key part of this project is building a real-time dataset. Instead of relying on pre-existing datasets, data is collected in actual conditions and stored in Google Sheets along with timestamps. This makes the dataset more realistic and suitable for practical applications.

The collected data is then used in the field of machine learning to train a model that can identify different human activities. The system is designed to classify whether a person is on the bed, sitting on a chair, present in the room, working in the kitchen, or outside the room. This classification is done using patterns observed in sensor readings such as distance changes, motion detection, and environmental conditions.

In addition to activity classification, the project also focuses on analyzing behavior patterns over time. By using graphs and time-based analysis, it becomes possible to observe daily routines, detect sleeping patterns, and understand activity distribution throughout the day. This makes the system useful not only for activity detection but also for behavioral analysis.

Overall, the area of work combines hardware and software technologies to build a smart monitoring system. It demonstrates how IoT devices and machine learning can work together to create an intelligent system capable of real-time data collection, analysis, and prediction.

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ABSTRACT

With the increasing advancement of smart technologies and the Internet of Things (IoT), there is a growing need for systems that can automatically monitor and understand human activities in real-time. This project presents a practical and cost-effective solution for activity monitoring by combining IoT-based data collection with machine learning techniques.

The system is developed using an ESP32 microcontroller, which acts as the central processing unit and communication device. It is connected to multiple sensors including a PIR sensor for detecting motion, an LDR sensor for measuring light intensity, a DHT22 sensor for monitoring temperature and humidity, and an ultrasonic sensor for measuring distance. These sensors continuously collect data from the environment and capture different aspects of human interaction within a room.

A major highlight of this project is the real-time data collection system that works independently without requiring a laptop. The ESP32 connects to a WiFi network and sends sensor data directly to Google Sheets using web-based APIs. Each data entry is stored along with a timestamp, which helps in building a structured dataset over time. This approach ensures that the collected data reflects real-world conditions, making it more suitable for practical analysis compared to pre-existing datasets.

The collected dataset is then processed and used to train a machine learning model capable of classifying different human activities. Based on patterns observed in motion, distance, light intensity, and environmental conditions, the model predicts whether a person is on the bed, sitting on a chair, present in the kitchen, or outside the room. The inclusion of multiple sensor inputs improves the reliability of predictions and allows better differentiation between activities.

In addition to activity classification, the project also focuses on analyzing behavioral patterns over time. Various graphs and visualizations are generated to understand daily routines, including hour-wise activity distribution, day-wise comparisons, and sensor variations. These visual insights help in identifying patterns such as sleeping behavior, active periods, and absence from the room.

Overall, this project demonstrates how simple hardware components and basic machine learning models can be effectively integrated to build a smart monitoring system. The system is easy to implement, low in cost, and capable of real-time operation. It can be further extended for applications such as smart home automation, elderly monitoring, health tracking, and anomaly detection, making it a useful solution in modern intelligent environments

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1. INTRODUCTION

1.1Background

In recent years, the rapid growth of Internet of Things (IoT) technology has made it possible to connect everyday devices and collect data in real time. This advancement has opened new opportunities in areas such as smart homes, healthcare monitoring, and automation systems. One of the important applications of IoT is human activity monitoring, where systems can observe and analyze the behavior of individuals based on sensor data.

Traditional activity monitoring systems often rely on cameras or wearable devices. While camera-based systems can provide accurate results, they raise serious privacy concerns and require complex processing. On the other hand, wearable devices depend on user cooperation and may not always be convenient for continuous monitoring. To overcome these limitations, sensor-based systems using embedded devices have emerged as a simple and effective alternative.

In this project, an IoT-based approach is used to monitor human activity inside a room using multiple sensors connected to an ESP32 microcontroller. The system collects environmental and positional data such as motion, light intensity, temperature, humidity, and distance. These parameters provide meaningful information about the presence and behavior of a person without requiring direct interaction or intrusion.

1.2 Motivation

The motivation behind this project comes from the need to develop a low-cost, real-time, and privacy-friendly system for activity monitoring. In many real-life scenarios such as smart homes, elderly care, and personal activity tracking, it is important to understand how a person behaves throughout the day.

Most existing solutions either depend on expensive hardware or are limited to controlled environments. Additionally, many systems rely on pre-collected datasets, which may not represent real-world conditions accurately. This project aims to address these challenges by collecting real-time data directly from sensors in a natural environment and using it to train a machine learning model.

Another important motivation is to bridge the gap between hardware and intelligent data analysis. Instead of just collecting sensor data, this project focuses on converting raw data into meaningful insights such as activity classification and behavioral patterns. This makes the system more practical and useful in real-world applications.

1.3 Scope of the Project

The scope of this project includes the design, development, and analysis of a complete IoT-based activity monitoring system. The system is capable of collecting real-time sensor data, storing it in a cloud platform, and using it for machine learning-based prediction.

The project mainly focuses on identifying different human activities inside a room. These activities include:

- Lying on the bed
- Sitting on a chair or table
- Being present in the kitchen
- Being outside the room

The system also includes time-based analysis to study daily routines and behavior patterns. By using timestamps and continuous data collection, it becomes possible to understand how activities change over time.

However, the scope is limited to sensor-based monitoring within a controlled indoor environment. The system does not include camera-based tracking or wearable devices. Future improvements can expand the system to include more sensors, advanced models, and larger environments.

1.4 Problem Statement

Understanding human activity using real-world data is a challenging task due to variations in behavior, environmental conditions, and sensor noise. It becomes even more difficult when the system needs to work continuously without manual intervention.

The main problem addressed in this project is to design a system that can:

- Collect real-time sensor data without using a computer
- Store data automatically in a structured format
- Use the collected data to train a machine learning model
- Accurately classify human activities based on sensor inputs
- Analyze behavior patterns over time

The challenge lies in integrating hardware and software components in such a way that the system remains simple, efficient, and reliable.

1.5 Objectives of the Project

The primary objectives of this project are as follows:

- To design a sensor-based system for real-time data collection
- To connect ESP32 with WiFi and transmit data to the cloud
- To create a structured dataset using Google Sheets
- To preprocess and analyze the collected data
- To train a machine learning model for activity classification
- To generate visual representations of behavior using graphs
- To evaluate the performance of the system in real-world conditions

1.6 Overview of the Proposed System

The proposed system consists of both hardware and software components working together. The sensors collect data and send it to the ESP32, which processes and transmits it via WiFi. The data is stored in Google Sheets and later used for machine learning.

The workflow of the system can be summarized as:

Sensors → ESP32 → WiFi → Google Sheets → Dataset → Machine Learning → Prediction → Graph Analysis

1.7 Significance of the Project

This project demonstrates how simple and affordable components can be used to build an intelligent system. It highlights the practical use of IoT and machine learning in solving real-world problems.

The system can be useful in:

- Smart home automation
- Health and activity monitoring
- Elderly care systems
- Behavior analysis

By focusing on real-time data collection and analysis, the project provides a strong foundation for future improvements and advanced applications.

2. BACKGROUND DETAILS AND LITERATURE REVIEW

2.1 Traditional Approaches

Earlier systems for monitoring human activity were mainly based on simple sensor readings or rule-based methods. In these systems, decisions were taken using fixed conditions. For example, if motion is detected, the system assumes that a person is present, and if no motion is detected, it assumes absence.

Although these methods are easy to implement, they are not very reliable in real-world situations. Human behavior is not always consistent, and environmental conditions can also change. Because of this, rule-based systems often fail to correctly identify activities, especially when multiple factors are involved.

Some systems also used individual sensors like motion sensors or light sensors separately. However, using only one type of sensor gives limited information and cannot clearly distinguish between different activities such as resting, sitting, or moving.

2.2 Sensor-Based Monitoring Systems

To improve the performance of activity monitoring, researchers started using multiple sensors together. These systems combine data from different sensors such as motion sensors, light sensors, temperature sensors, and distance sensors.

Using multiple sensors provides more information about the environment and helps in better understanding of human activity. For example, motion combined with distance can indicate whether a person is moving or staying in one position, while light intensity can give an idea about day or night conditions.

However, even with multiple sensors, early systems still relied on simple logic and fixed thresholds. This limited their ability to handle variations in real-world conditions. As a result, these systems were not always accurate in predicting activities.

2.3 Machine Learning-Based Approaches

With the advancement of machine learning, activity recognition systems have improved significantly. Instead of using fixed rules, machine learning models learn patterns from data and use those patterns to make predictions.

Different models such as decision trees, support vector machines, and random forest classifiers have been used for activity classification. These models can handle multiple inputs at the same time and are better at identifying complex patterns in data.

Machine learning also allows the system to improve over time as more data is collected. This makes the system more flexible and suitable for real-world applications.

2.4 IoT-Based Real-Time Systems

In recent years, IoT-based systems have become popular for real-time data collection and monitoring. Microcontrollers like ESP32 allow sensors to be connected and data to be transmitted over the internet.

These systems can continuously collect data without requiring manual intervention. Cloud platforms such as Google Sheets or databases can be used to store this data along with timestamps. This makes it possible to analyze behavior over long periods of time.

IoT-based systems are also cost-effective and easy to implement, which makes them suitable for practical applications like smart homes and monitoring systems.

2.5 Summary

From the study of existing systems, it can be observed that:

- Rule-based systems are simple but not very accurate
- Single sensor systems provide limited information
- Multi-sensor systems improve performance but still need better analysis
- Machine learning improves accuracy by learning from data
- IoT enables real-time data collection and storage

Based on these observations, this project combines multiple sensors with machine learning and IoT to create a system that can accurately detect human activity and analyze behavior in real time.

3. PROBLEM DEFINITION AND OBJECTIVES

3.1 Problem Definition

The main problem addressed in this project is the accurate identification of human activity inside a room using real-time sensor data. In a practical environment, it is difficult to determine what a person is doing based only on simple sensor readings because human behavior can vary and sensor values are not always consistent.

Another challenge is that most basic systems rely on fixed conditions or thresholds, which are not suitable for real-world situations. For example, motion detection alone cannot differentiate between activities like sitting, resting, or moving slowly. Similarly, distance or light values by themselves cannot provide a complete understanding of the situation.

In addition to this, collecting real-time data continuously without using a laptop is also a challenge. The system must be able to work independently, store data properly, and handle variations in sensor readings.

Therefore, the problem is to design a system that can:

- Collect real-time data from multiple sensors
- Store the data in a structured format automatically
- Use the collected data to identify different human activities
- Work efficiently in real-world conditions without manual intervention

3.2 Objectives

The main objectives of this project are:

- To design and develop a system using ESP32 and sensors for real-time data collection
- To connect the system to WiFi and send data to a cloud platform (Google Sheets)
- To generate a dataset based on actual sensor readings collected over time
- To preprocess the collected data and prepare it for analysis
- To train a machine learning model using the dataset
- To classify human activity into different categories such as bed, chair, kitchen, and outside
- To analyze behavior patterns using time-based data
- To represent the results using graphs for better understanding

3.3 Activity Classes

The system is designed to identify the following activities:

Activity	Description
Bed	Person lying or resting on bed
Chair	Person sitting on chair or table
Kitchen	Person present near kitchen area
Outside	Person not present in room

4. PROPOSED METHODOLOGY AND WORK DESCRIPTION

4.1 System Overview

The proposed system is designed to monitor human activity inside a room using multiple sensors connected to an ESP32 microcontroller. The system works by collecting real-time data from the environment and analyzing it using machine learning techniques.

The overall system consists of three main parts:

- **Sensor Unit** – collects data such as motion, light, temperature, humidity, and distance
- **Processing Unit (ESP32)** – reads sensor values and sends data through WiFi
- **Cloud and Analysis Unit** – stores data and performs machine learning analysis

All these components work together to create a continuous data collection and prediction system. The ESP32 acts as the central controller, which connects all sensors and handles communication with the cloud.

4.2 Data Collection

The system collects data using the following sensors:

- PIR sensor for motion detection
- LDR sensor for light intensity
- DHT22 sensor for temperature and humidity
- Ultrasonic sensor for distance measurement

These sensors continuously generate data at regular time intervals. Each reading includes multiple parameters, which together provide a better understanding of human activity.

The ESP32 reads these sensor values and sends them to Google Sheets using WiFi. Each data entry is stored along with the current timestamp, which helps in tracking activities over time.

Unlike many systems that use pre-existing datasets, this project focuses on collecting data in real conditions. This makes the dataset more realistic and suitable for training the model.

4.3 Data Storage

The collected data is stored in Google Sheets using a web-based API. This allows the system to store data automatically without manual effort.

Each row of the dataset contains:

- Time
- Motion
- Light intensity
- Temperature
- Humidity
- Distance

This structured format makes it easy to analyze and use the data for machine learning.

4.4 Data Preprocessing

Before using the data for machine learning, it is necessary to clean and prepare it. This step ensures that the model receives accurate and meaningful input.

The preprocessing steps include:

- Removing missing or incorrect values
- Converting time into useful features such as hour and day
- Organizing the data into proper format
- Normalizing values if required

These steps help in improving the performance of the machine learning model.

4.5 Machine Learning Model

In this project, a machine learning model is used to classify human activity based on sensor data. The model is trained using the dataset collected from real-time sensor readings.

A Random Forest classifier is used because it works well with multiple input features and provides good accuracy for classification problems.

The model takes the following inputs:

- Motion
- Light intensity
- Temperature
- Humidity
- Distance
- Time (hour)

Based on these inputs, the model predicts the activity of the person.

4.6 Activity Classification

The system classifies human activity into different categories such as:

- Bed
- Chair
- Kitchen
- Outside

The classification is based on patterns observed in the data. For example:

- Stable distance with no motion indicates sleeping or resting
- Moderate distance with motion indicates sitting
- High activity and changing distance indicates movement

By combining multiple sensor values, the system can make more accurate predictions.

4.7 Data Visualization

To understand the behavior patterns, the collected data is visualized using different graphs. These graphs help in analyzing how activities change over time.

The following graphs are used:

- Daily routine graph (hours spent in activities)
- Hour vs day heatmap
- Distance vs time graph
- Temperature and humidity variation

These visualizations make it easier to identify patterns such as sleep cycles, active hours, and absence from the room.

4.8 Overall Working

The complete working of the system can be summarized as follows:

Sensors → ESP32 → WiFi → Google Sheets → Dataset → Machine Learning Model → Activity Prediction → Graph Analysis

5. PROPOSED ALGORITHMS

5.1 Data Collection Algorithm

The system collects data from multiple sensors connected to the ESP32. The process starts when the ESP32 is powered on and connected to a WiFi network. After establishing the connection, the sensors begin to read values continuously at regular time intervals.

At each step, the ESP32 reads input values from the motion sensor, light sensor, temperature and humidity sensor, and the ultrasonic sensor. These values are combined into a single data record. Along with the sensor readings, the current time is also included to keep track of when the data is collected.

Once the data is prepared, it is sent to Google Sheets using a web request. This process is repeated continuously, which helps in building a dataset over time without any manual intervention.

5.2 Data Processing Algorithm

After collecting the data, it is processed to make it suitable for machine learning. The first step is to check for any missing or incorrect values and remove them if necessary. Then, the time information is converted into useful features such as hour and day.

The data is then organized in a structured format so that it can be easily used for training the model. If required, normalization is applied to ensure that all values are within a suitable range.

This step ensures that the dataset is clean, consistent, and ready for analysis.

5.3 Training Algorithm

The machine learning model is trained using the processed dataset. The dataset is divided into two parts: training data and testing data. The training data is used to teach the model, while the testing data is used to evaluate its performance.

A Random Forest classifier is used for training. The model takes multiple input features such as motion, light intensity, temperature, humidity, distance, and time. During training, the model learns patterns that correspond to different activities.

After training, the model is tested using unseen data to check its accuracy. The trained model is then saved for future predictions.

5.4 Prediction Algorithm

Once the model is trained, it can be used to predict human activity. New sensor data is given as input to the model, and based on the learned patterns, the model predicts the activity category.

The prediction process involves taking sensor readings, converting them into the required format, and passing them to the model. The model then outputs one of the predefined activity classes such as bed, chair, kitchen, or outside.

This process can be repeated continuously to provide real-time predictions.

5.5 Overall Algorithm

The complete working of the system can be summarized in the following steps:

Step 1: Initialize ESP32 and connect to WiFi
Step 2: Read data from sensors
Step 3: Add timestamp to data
Step 4: Send data to Google Sheets
Step 5: Store and organize dataset
Step 6: Preprocess the data
Step 7: Train machine learning model
Step 8: Predict activity using trained model
Step 9: Analyze results using graphs

6. PROPOSED FLOWCHART / BLOCK DIAGRAM / DFD

6.1 System Architecture

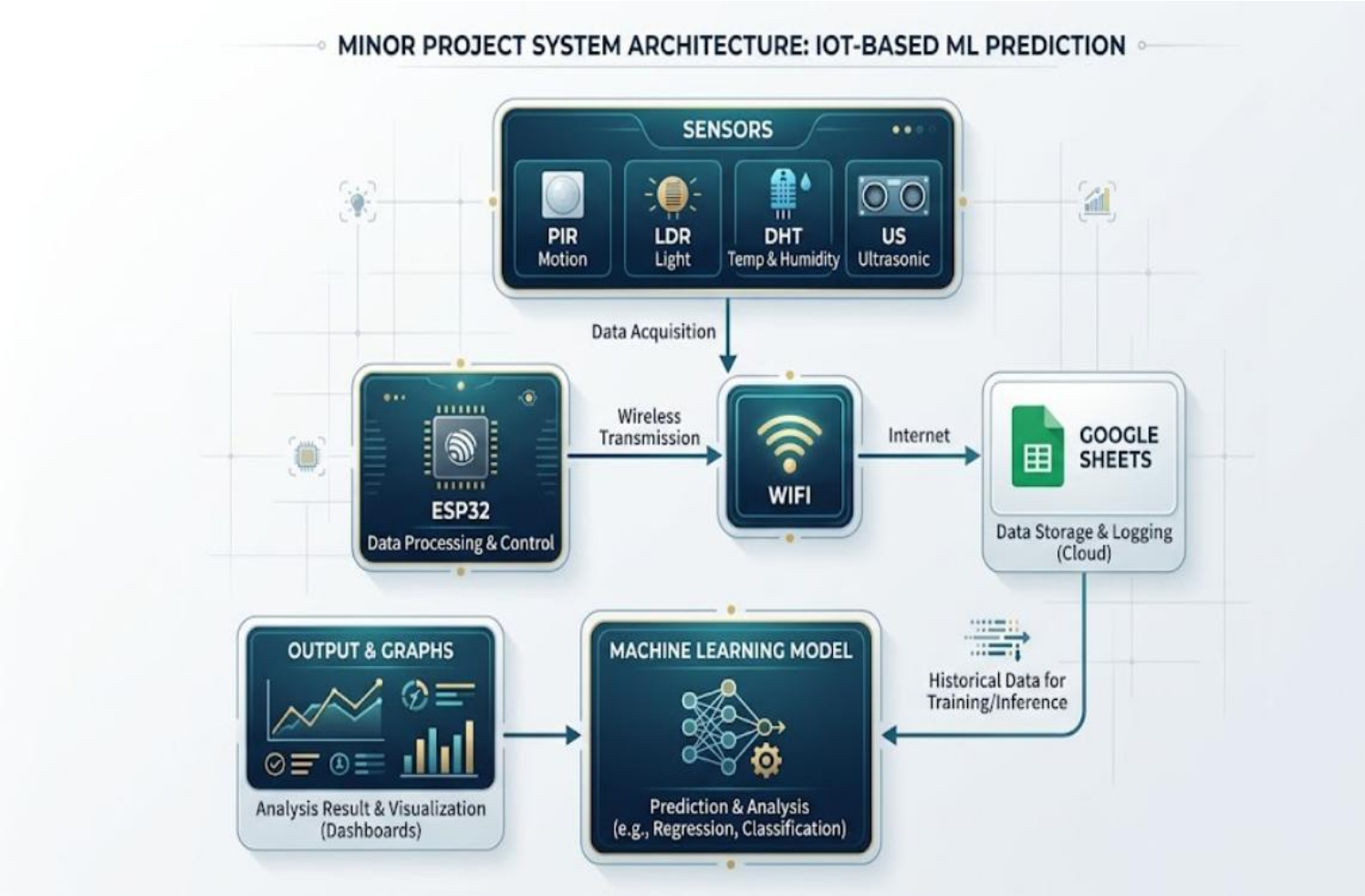
The proposed system is designed as a simple and efficient IoT-based architecture where different components work together to collect, store, and analyze data.

The system consists of three main parts:

- **Input Layer (Sensors)** – collects real-time data
- **Processing Layer (ESP32)** – processes and transmits data
- **Analysis Layer (Cloud + Machine Learning)** – stores and analyzes data

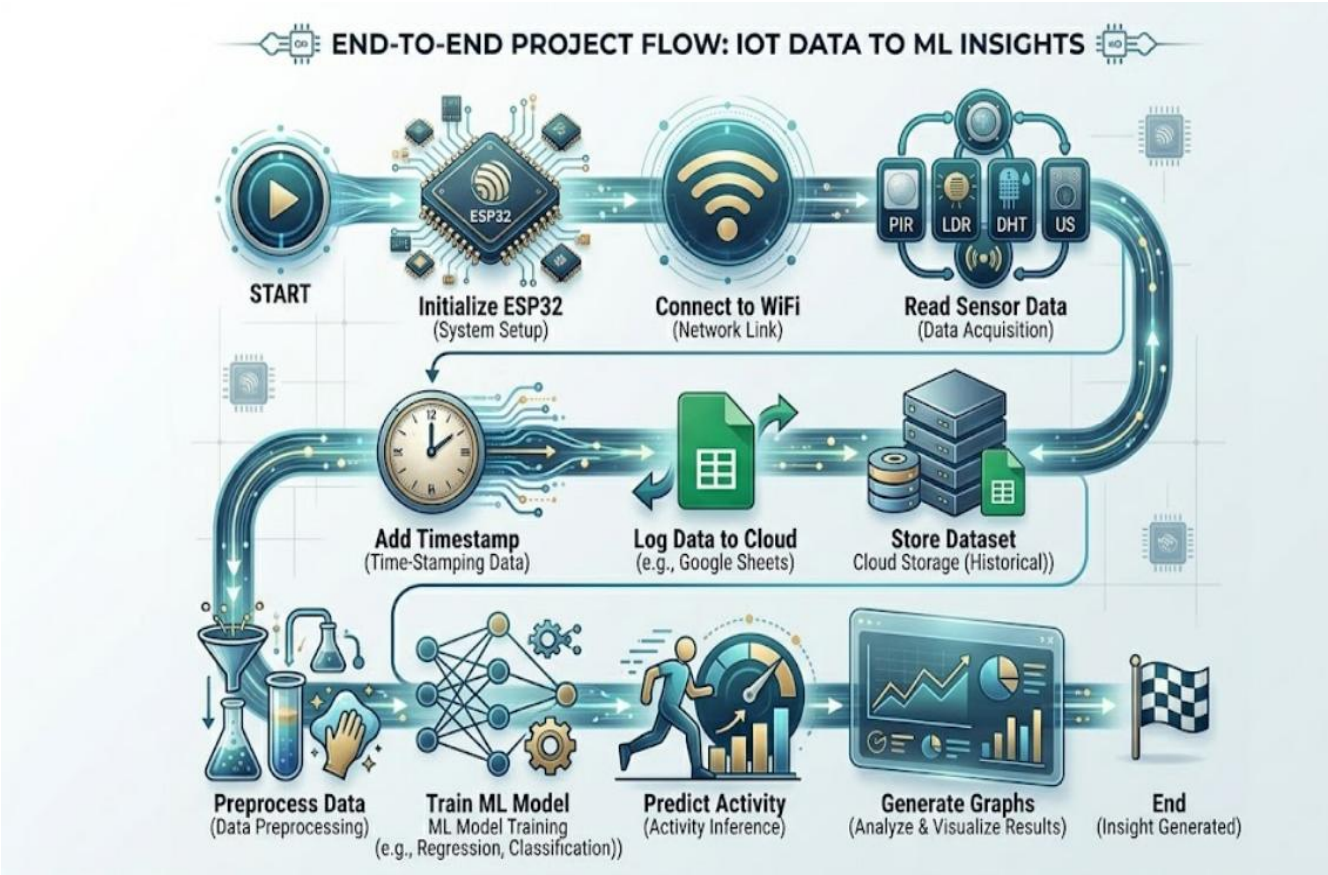
The sensors continuously send data to the ESP32. The ESP32 processes the data and transmits it through WiFi to Google Sheets. This data is later used for machine learning and analysis.

6.2 Block Diagram



6.3 Flowchart

This shows step-by-step working of system



6.4 Data Flow Description (DFD)

The data flow in the system starts from the sensors, which collect real-time data from the environment. This data is sent to the ESP32, where it is processed and formatted.

The ESP32 then sends the data through WiFi to Google Sheets, where it is stored in a structured format. The stored data is later used as input for the machine learning model.

The model processes the data and generates predictions about human activity. Finally, the results are visualized using graphs to understand behavior patterns.

7. TOOLS AND TECHNOLOGY USED (IMPLEMENTATION)

7.1 Hardware Components

The hardware used in this project is simple and cost-effective, but when combined, it creates a powerful system for real-time data collection. Each component plays a specific role in understanding the environment and human activity.

ESP32 Microcontroller

The ESP32 is the main controller of the system. It is responsible for reading data from all sensors, processing that data, and sending it to the cloud using WiFi. One of the main advantages of ESP32 is that it has built-in WiFi, which removes the need for any additional communication module.

It operates at **3.3V logic level** and supports both digital and analog inputs, making it suitable for connecting multiple sensors at the same time. In this project, it acts as the central unit that connects all hardware and software components together.

PIR Sensor (Motion Sensor)

The PIR (Passive Infrared) sensor is used to detect human motion. It works by sensing changes in infrared radiation emitted by the human body.

- **Output Type:** Digital (0 or 1)
- **Detection Range:** Approximately **3 to 7 meters**
- **Detection Angle:** Around **120 degrees**

When a person moves in front of the sensor, it gives a HIGH signal (1), otherwise it remains LOW (0). This helps in identifying whether there is activity in the room.

LDR Sensor (Light Sensor)

The LDR (Light Dependent Resistor) is used to measure light intensity. It works based on resistance change — when light increases, resistance decreases.

- **Output Type:** Analog (0–4095 for ESP32)
- **Typical Range Used in Project:**
    - Light ON → around **960 to 975**
    - Light OFF → around **1065 to 1089**

This sensor helps in identifying whether the room light is ON or OFF, which indirectly gives information about user activity (awake or sleeping).

DHT22 Sensor (Temperature & Humidity)

The DHT22 sensor is used to measure environmental conditions such as temperature and humidity. It provides more accurate readings compared to basic sensors.

- **Temperature Range: -40°C to 80°C**
- **Accuracy:** ±0.5°C
- **Humidity Range: 0% to 100%**
- **Accuracy:** ±2–5%

In this project, temperature and humidity values help in understanding environmental changes and add extra information for machine learning analysis.

Ultrasonic Sensor (HC-SR04)

The ultrasonic sensor is used to measure distance by sending sound waves and calculating the time taken for the echo to return.

- **Range: 2 cm to 400 cm**
- **Working Range (practical): 20 cm to 300 cm**
- **Accuracy:** ±3 mm

In this project, distance plays a very important role. It helps in identifying whether a person is on the bed, sitting on a chair, or not present near the sensor.

For example:

Bed	Distance remains within fixed range
Chair	Moderate and changing distance
Outside	Large distance

Breadboard and Jumper Wires

A breadboard is used to connect all components without soldering. Jumper wires are used to make connections between ESP32 and sensors. This setup makes the system flexible and easy to modify during testing.

7.2 Software Tools

The software tools used in this project are responsible for programming, data storage, and analysis.

Arduino IDE

Arduino IDE is used to write and upload code to the ESP32. The code handles:

- Sensor reading
- WiFi connection

- Data transmission

It provides a simple interface and is widely used for embedded system projects.

Google Sheets

Google Sheets is used as a cloud-based database in this project. The ESP32 sends data directly to Google Sheets, where it is stored in rows along with timestamps.

This approach is very useful because:

- No need for local storage
- Data is accessible from anywhere
- Easy to analyze and export

Google Apps Script

Google Apps Script is used to create a web API that receives data from ESP32. Whenever ESP32 sends data, this script automatically adds a new row in Google Sheets.

This makes the entire system automatic and eliminates manual data entry.

7.3 Programming and Libraries

Arduino Programming (C/C++)

The ESP32 is programmed using C/C++ in Arduino IDE. The program includes:

- Sensor interfacing
- Data processing
- HTTP requests for sending data

Python

Python is used for data analysis and machine learning. It helps in:

- Cleaning data
- Training model
- Generating graphs

Libraries Used

WiFi.h	for internet connection
HTTPClient.h	for sending data
DHT.h	for temperature & humidity sensor
Pandas	for data handling
Scikit-learn	for ML model
Matplotlib	for visualization

7.4 Machine Learning Tools

Machine learning is used to classify human activity based on sensor data.

Scikit-learn

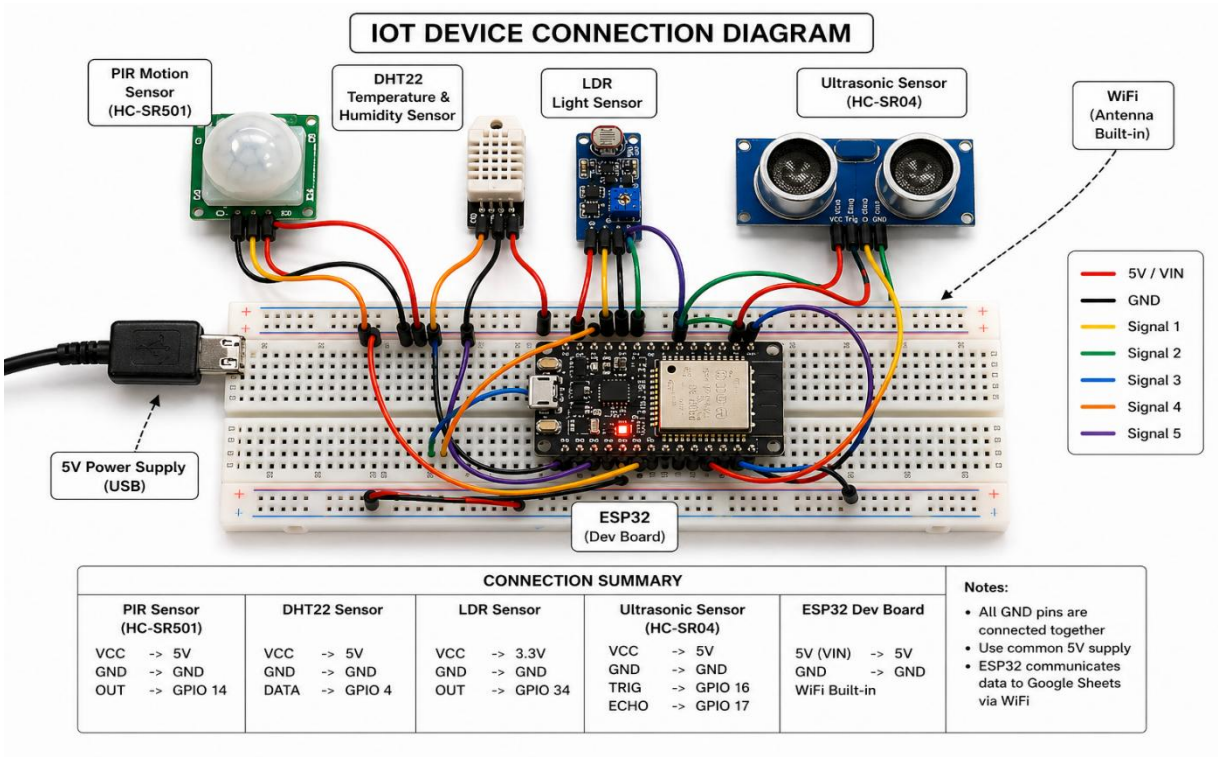
Scikit-learn is used to implement the Random Forest model. It is simple, efficient, and suitable for classification problems.

Data Visualization

Graphs are generated using Matplotlib to understand behavior patterns such as:

- Daily routine
- Activity distribution
- Sensor variation

7.6 Hardware Implementation



Sensor Name	VCC	GND	Signal Pin	ESP32 GPIO
PIR Sensor (HC-SR501)	5V	GND	OUT	GPIO 13
DHT22 Sensor	5V	GND	DATA	GPIO 4
LDR Sensor	3.3V	GND	Analog OUT	GPIO 34

Sensor Name	VCC	GND	Signal Pin	ESP32 GPIO
Ultrasonic (HC-SR04)	5V	GND	TRIG	GPIO 5
Ultrasonic (HC-SR04)	5V	GND	ECHO	GPIO 18
ESP32 Dev Board	5V (VIN)	GND	—	—

The hardware setup of the project is built using an ESP32 microcontroller and multiple sensors connected through a breadboard. Each sensor is connected carefully according to its functionality.

The PIR sensor is connected to detect motion, while the LDR is connected using a voltage divider circuit to measure light intensity. The DHT22 sensor is used to measure temperature and humidity, and the ultrasonic sensor is used to calculate distance.

All components share a common ground, and the ESP32 is powered using a USB cable connected to a power bank. This allows the system to work independently without a laptop.

7.7 Software Implementation

The software part of the project is developed using Arduino IDE. The ESP32 is programmed to read sensor data and send it to Google Sheets using WiFi.

At the beginning, all sensors are initialized and WiFi connection is established. After that, the system continuously reads sensor values and sends them to the cloud using HTTP requests.

This process runs in a loop, enabling continuous and automatic data collection.

7.8 Data Collection

The system collects real-time data from sensors at regular intervals. Each data entry includes multiple parameters such as motion, light intensity, temperature, humidity, and distance.

The data is stored in Google Sheets along with timestamps. This allows the dataset to grow over time and capture different conditions such as day, night, and activity changes.

The use of a power bank makes the system portable and suitable for long-term data collection.

7.9 Dataset Preparation

After collecting the data, it is prepared for machine learning. The dataset is cleaned to remove any incorrect or missing values.

Time-related information is converted into useful features such as hour and day. The data is then labeled based on activities such as bed, chair, kitchen, and outside.

All datasets are combined into a single dataset for training.

7.10 Machine Learning Implementation

The machine learning model is implemented using Python. The dataset is divided into training and testing sets.

A Random Forest classifier is used to train the model. The model learns patterns from the data and uses them to classify activities.

After training, the model is tested using new data to evaluate its accuracy.

7.11 Prediction and Analysis

Once the model is trained, it is used to predict human activity based on sensor inputs.

The system classifies activities into categories such as bed, chair, kitchen, and outside. The predictions are then analyzed using graphs to understand behavior patterns.

Graphs such as daily routine and sensor variation help in visualizing the results.

7.12 Challenges Faced

During the implementation of the project, some challenges were encountered:

- Incorrect wiring of LDR caused constant values initially
- Delay in data transmission affected motion detection
- Noise in sensor readings
- Difficulty in collecting consistent real-world data

These challenges were resolved by improving connections, optimizing code, and collecting more data.

8. RESULT DISCUSSION AND ANALYSIS

8.1 Overview of Results

The system successfully collected real-time data using sensors and stored it in Google Sheets. This data was then used to train a machine learning model for activity classification.

Different graphs were generated to analyze behavior patterns across multiple days. These graphs clearly show variations in temperature, humidity, light intensity, and activity over time.

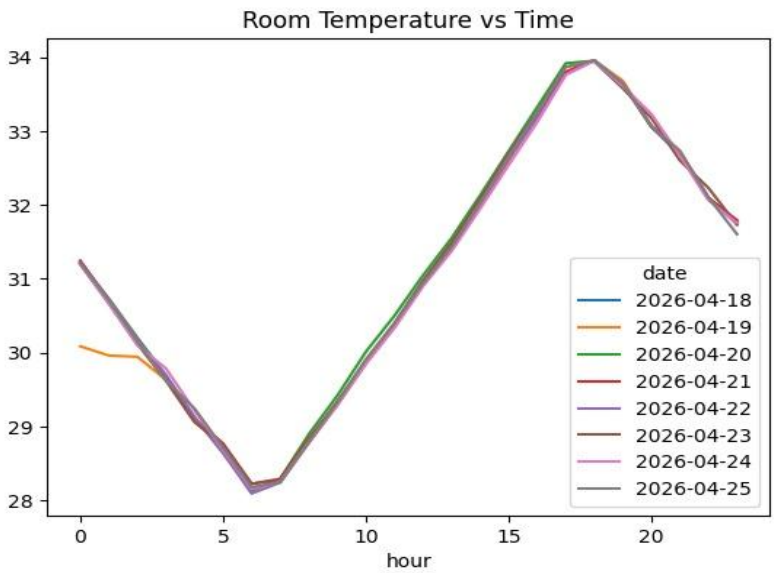
8.2 Environmental Analysis (Room Data)

Room Temperature vs Time

From the graph, it can be observed that temperature follows a natural daily pattern:

- Temperature is lowest in early morning (around **28–29°C**)
- It gradually increases during the day
- Peaks in evening (around **33–34°C**)
- Decreases again at night

This shows that the sensor readings are realistic and follow expected environmental trends.

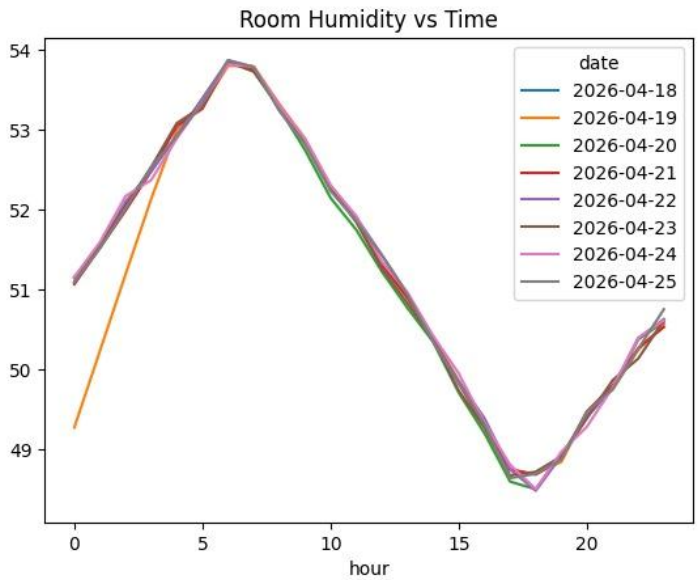


Room Humidity vs Time

The humidity graph shows an inverse trend compared to temperature:

- Higher humidity in early morning (**around 53–54%**)
- Gradual decrease during the day
- Lowest in evening (**around 48–49%**)
- Slight increase again at night

This indicates that humidity is affected by temperature changes, which is expected in real conditions.



Room Light vs Time

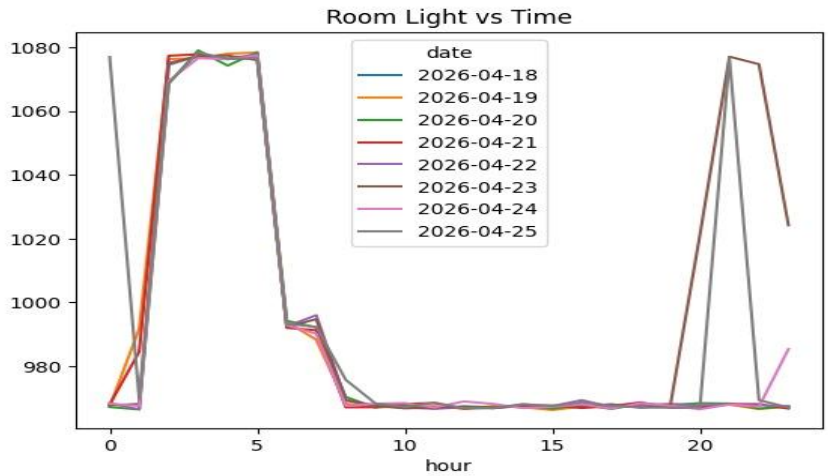
The light intensity graph shows clear patterns:

- High values (~1060–1080) when lights are OFF or no direct light
- Lower values (~965–980) when light is ON

It can also be observed that:

- Light changes sharply during morning and night
- Some variations occur due to human activity

This helps in identifying whether the person is active or sleeping.



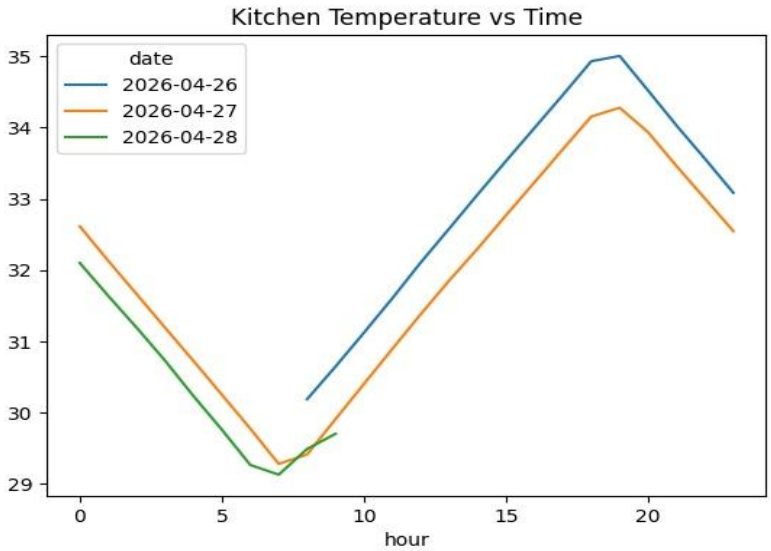
8.3 Kitchen Data Analysis

Kitchen Temperature vs Time

The kitchen temperature graph shows:

- Lower temperature in morning (~29°C)
- Gradual increase during day
- Peak around afternoon (~34–35°C)

This increase is slightly higher compared to room temperature, which may be due to cooking activities.

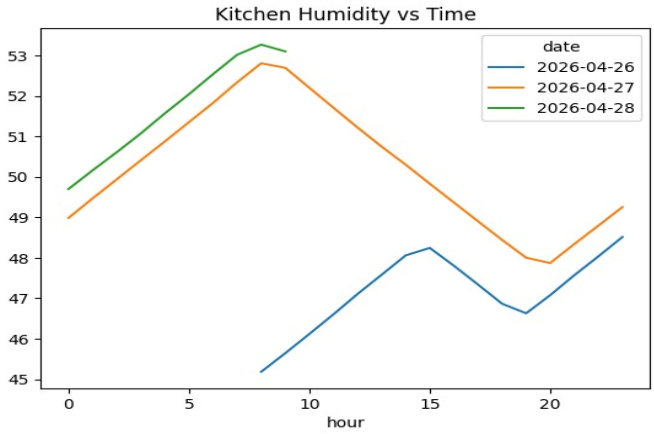


Kitchen Humidity vs Time

The humidity in kitchen shows:

- Increasing trend in morning
- Peak around mid-day (~53%)
- Gradual decrease in evening

This behavior is realistic because cooking activities can increase humidity levels.

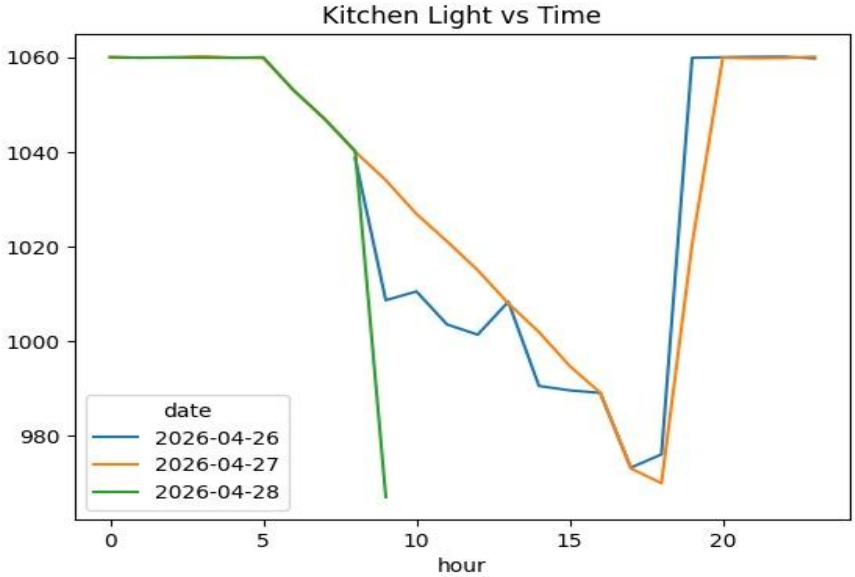


Kitchen Light vs Time

The kitchen light graph shows:

- High intensity (~1060) when lights are OFF
- Gradual decrease when light is ON
- Sharp variations indicating activity

This helps in identifying presence in the kitchen.



8.4 Activity-Based Analysis

Activity vs Hour Graph

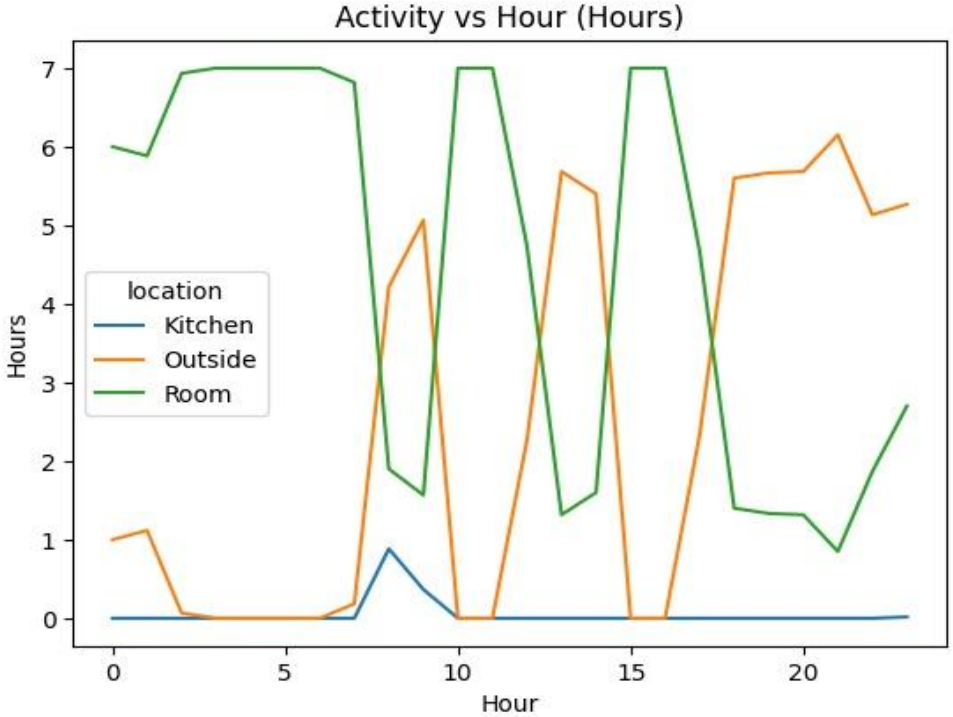
This graph shows how time is distributed among:

- Room
- Kitchen
- Outside

Observations:

- Maximum time is spent in the room
- Kitchen usage is limited and occurs at specific times
- Outside activity increases during evening hours

This clearly reflects a realistic daily routine.

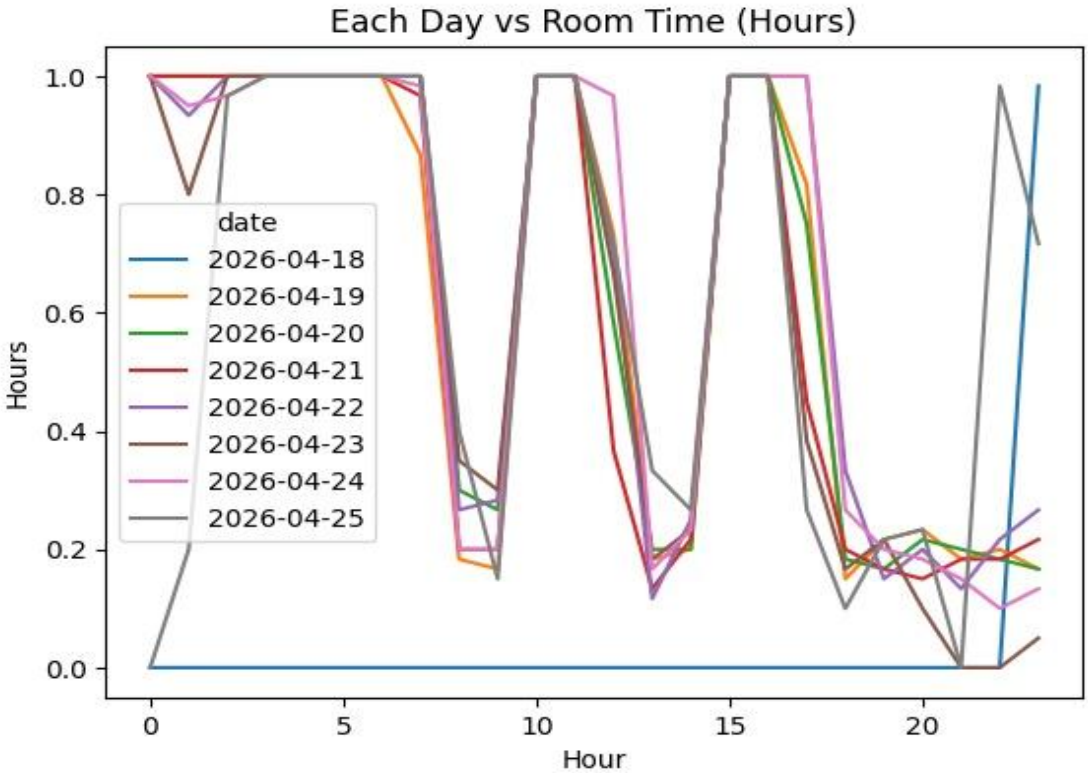


Room Time vs Hour

This graph shows:

- High presence in room during night and early morning
- Decrease in room time during day
- Sudden drops indicating absence

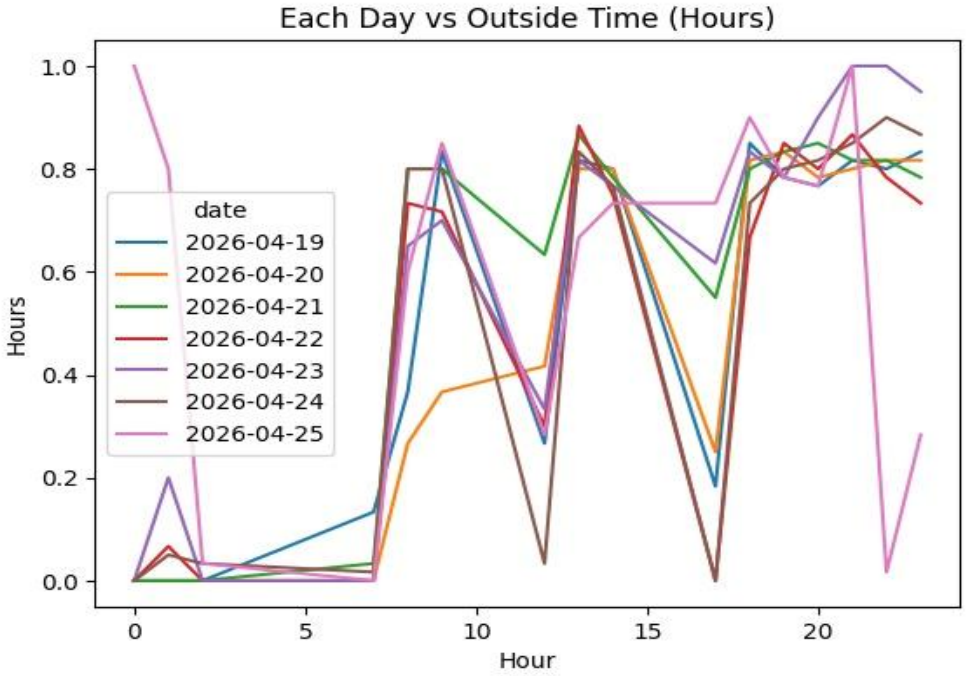
This aligns with normal human behavior.



Outside Time vs Hour

From this graph:

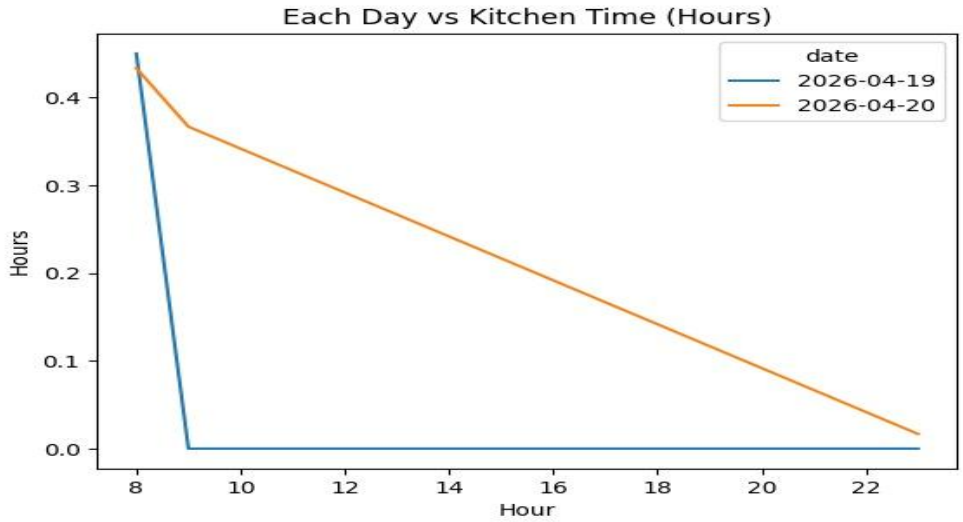
- Outside activity is mostly during day and evening
- Minimal presence outside during night
- Some variations based on daily schedule



Kitchen Time vs Hour

This graph shows:

- Short duration of kitchen usage
- Mostly during morning or meal times
- Very limited usage compared to room



8.5 Machine Learning Performance

The machine learning model was trained using the collected dataset. After testing, the model achieved an accuracy of approximately:

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76% - 79%
```

The model was able to correctly classify most activities such as:

- Bed (low motion, stable distance)
- Chair (moderate motion, varying distance)
- Kitchen (specific time + activity pattern)
- Outside (no sensor activity in room)

8.6 Key Observations

From all the graphs and results, the following observations can be made:

- Temperature and humidity follow natural daily patterns
- Light intensity clearly indicates active and inactive periods
- Distance and motion are the most important features for activity detection
- The system is able to capture realistic human behavior
- Activity distribution graphs match expected daily routine

8.7 Limitations

Although the system performs well, some limitations were observed:

- Sensor noise can affect readings
- Similar activities (bed vs chair) may sometimes overlap
- Accuracy depends on quality and amount of data
- Real-time delay may affect fast motion detection

8.8 Final Result

The system successfully integrates IoT and machine learning to monitor and analyze human activity. It provides:

- Real-time data collection
- Accurate activity prediction
- Meaningful visualization through graphs

Overall, the project meets its objective and demonstrates a practical implementation of smart activity monitoring.

9. CONCLUSION AND FUTURE SCOPE

9.1 Conclusion

In this project, a complete system was developed to monitor and analyze human activity using IoT and machine learning. The system uses multiple sensors such as PIR, LDR, DHT22, and an ultrasonic sensor to collect real-time data related to motion, light intensity, temperature, humidity, and distance.

The ESP32 microcontroller plays a central role by collecting data from sensors and sending it to Google Sheets through WiFi. This allows continuous data collection without the need for a laptop. The collected dataset reflects real-life conditions and includes variations based on time, activity, and environment.

Using this dataset, a machine learning model was trained to classify different activities such as being on the bed, sitting on a chair, being in the kitchen, or being outside the room. The model was able to achieve good accuracy and showed reliable performance in most cases.

The graphical analysis further helped in understanding daily behavior patterns, environmental changes, and activity distribution over time. The results demonstrate that sensor data can be effectively used to detect and analyze human activities in a smart environment.

Overall, the project successfully integrates hardware, software, and machine learning into a single system and achieves its objective of intelligent activity monitoring.

9.2 Future Scope

Although the system performs well, there are several ways in which it can be improved and expanded in the future.

One possible improvement is to use more advanced sensors such as camera-based systems or wearable devices to increase accuracy. Additional sensors like gas sensors or sound sensors can also be added to enhance activity detection.

The machine learning model can be improved by using larger datasets and more advanced algorithms such as deep learning. This will help in reducing errors and improving prediction accuracy, especially for similar activities.

Another enhancement could be the development of a mobile or web application to display real-time data and predictions. This would make the system more user-friendly and accessible.

The system can also be extended for real-world applications such as:

- Smart home automation
- Health monitoring for elderly people
- Security and surveillance systems
- In addition, real-time alerts can be added to notify users in case of unusual activity or emergencies.

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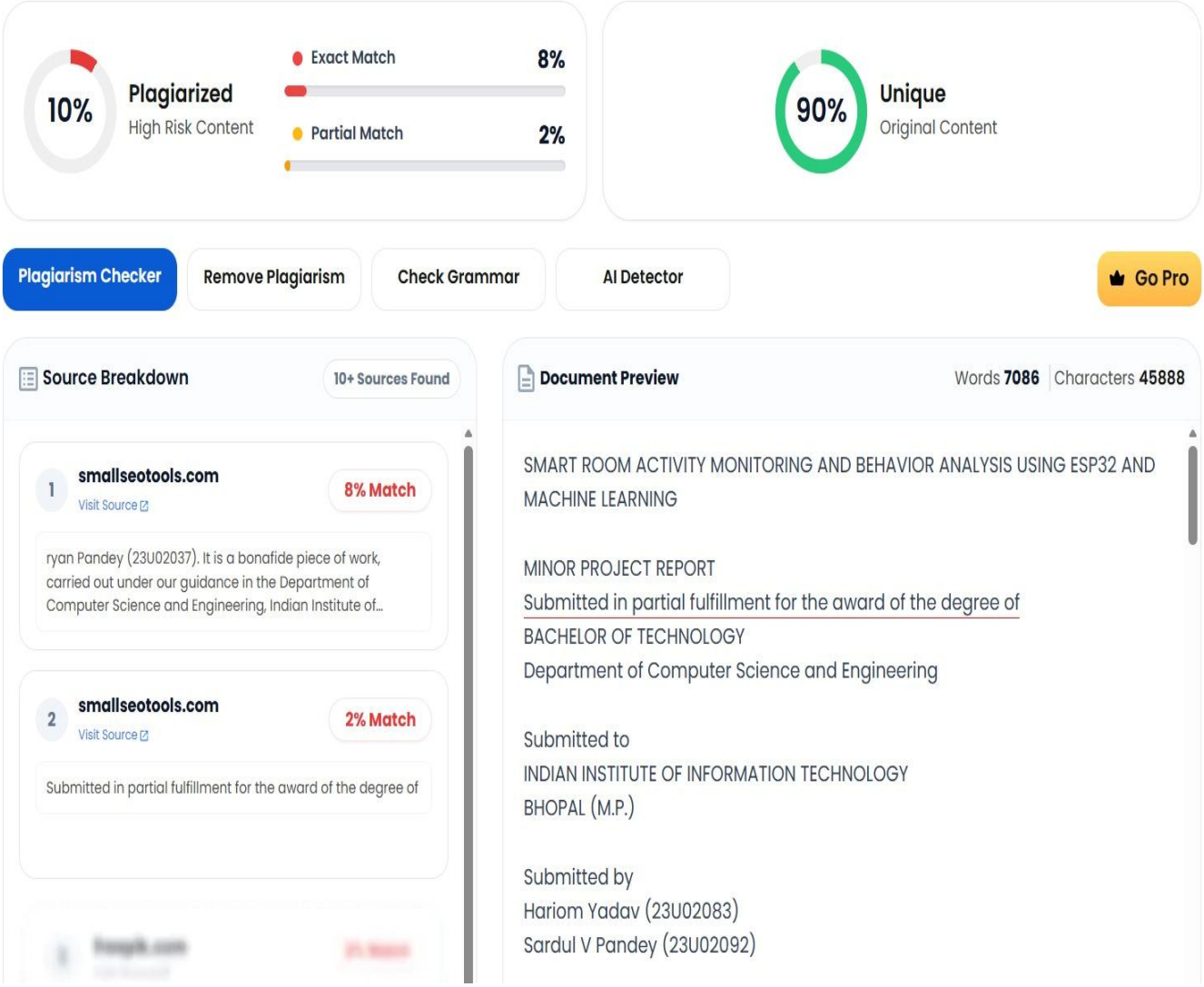
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